INCH-POUND

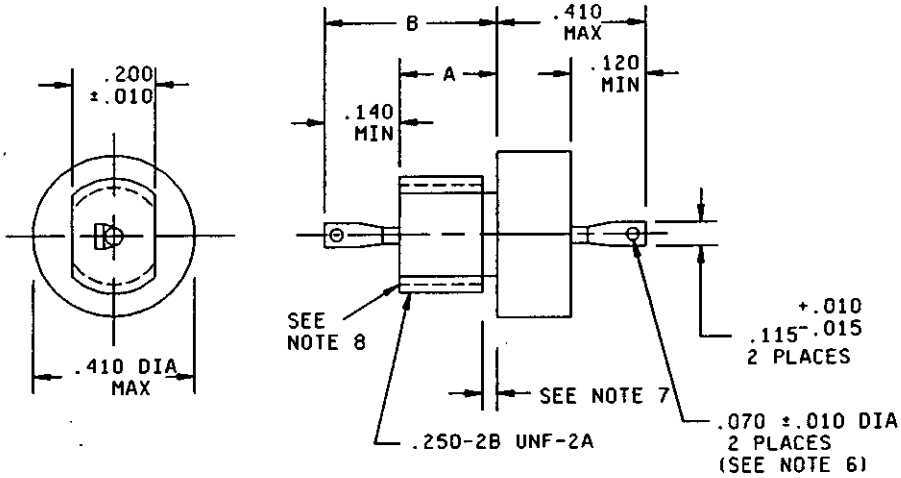
MIL-F-28861/1C
1 December 1993
SUPERSEDING
MIL-F-28861/1B
25 May 1990

MILITARY SPECIFICATION SHEET

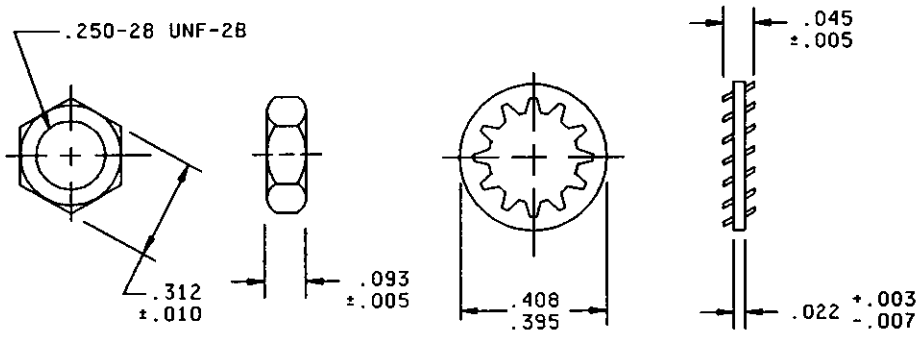
# FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLES FS10 AND FS11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.




STYLES FS10 AND FS11




HEX NUT

LOCKWASHER

FIGURE 1. Case and hardware dimensions and circuit diagrams.

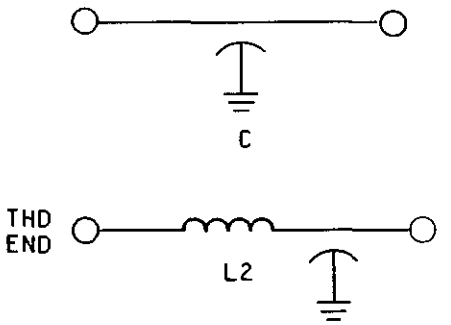
Ⓒ denotes changes



DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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| Dash number | A dimension (±.010) | B dimension (max) |
|---|---|---|
| 001 thru 010 | .187 | .357 |
| 011 thru 020 | .312 | .482 |




CIRCUIT DIAGRAMS

| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .140 | 3.56 |
| .005 | 0.13 | .150 | 3.81 |
| .007 | 0.18 | .187 | 4.75 |
| .010 | 0.25 | .200 | 5.08 |
| .015 | 0.38 | .250 | 6.35 |
| .022 | 0.56 | .312 | 7.92 |
| .045 | 1.14 | .357 | 9.07 |
| .070 | 1.78 | .395 | 10.03 |
| .093 | 2.36 | .408 | 10.36 |
| .115 | 2.92 | .410 | 10.41 |
| .120 | 3.05 | .482 | 12.24 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter with the symbol "L" or the circuit diagram shall be marked on the case.
6. Optional slot may be supplied, .050 ±.010 inch (1.27 ±0.25 mm) x .070 ±.010 inch (1.78 ±0.25 mm).
7. Imperfect thread or undercut optional .050 inch (1.27 mm) maximum.
8. One imperfect thread allowed .035 inch (0.89 mm) maximum.
9. Recommended mounting torque: 44 ounce-inch ±4 ounce-inch.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.



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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 5 grams maximum.

Case finish:

(C) Class B filters, T only: Tin plated or tin-lead plated.

(C) Class S filters, T or G only: Tin plated, tin-lead plated, or gold plated, in accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 15 amperes.

Capacitance: See table I.

Dissipation factor: 3 percent maximum.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

*Insulation resistance:*

At +25°C: 1,000 megohms minimum or 1,000 megohm-microfarads minimum, whichever is less.

At +125°C: 100 megohms minimum or 100 megohm-microfarads minimum, whichever is less.

Insertion loss: See table I.

Voltage drop: 0.12 V dc maximum.

*DC resistance: 0.008 ohm maximum.*

Seal: In accordance with MIL-F-28861. Leakage rate for class S filters shall not exceed 1 x $10^{-7}$ atm $cm^3/s$.

Temperature rise: +25°C maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Product assurance level: In accordance with table I.

Part or Identifying Number (PIN): The PIN shall be as follows:

M28861/01 - XXX X X

| XXX | X | X |
|---|---|---|
| Dash no. in accordance with figure 1 and table I | Case finish | Product assurance level in accordance with MIL-F-28861 and table I |



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TABLE I. Electrical characteristics.

| Dash number | Product assurance level Class B | Product assurance level Class S | Style | Circuit | Rated voltage DC volts | Rated voltage AC 1/ volts | Minimum capacitance (μF) | Minimum insertion loss (dB) in accordance 2/ 3/ with MIL-STD-220 At +25°C 30 kHz | 150 kHz | 300 kHz | 1 MHz | 10 MHz | 100 MHz | 1 GHz | At -55°C and +125°C 30 kHz | 150 kHz | 300 kHz | 1 MHz | 10 MHz | 100 MHz | 1 GHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 001, 011 | X | --- | FS10 | L2 | 50 | --- | 1.2 | 15 | 28 | 33 | 40 | 40 | 70 | 70 | 13 | 26 | 31 | 38 | 38 | 70 | 70 |
| 002, 012 | X | --- | FS10 | C | 50 | --- | 1.2 | 15 | 28 | 33 | 40 | 40 | 70 | 70 | 13 | 26 | 31 | 38 | 38 | 70 | 70 |
| 003, 013 | X | X | FS11 | L2 | 70 | --- | 0.7 | 10 | 24 | 30 | 40 | 40 | 64 | 70 | 8 | 22 | 28 | 38 | 38 | 64 | 70 |
| 004, 014 | X | X | FS11 | C | 70 | --- | 0.7 | 10 | 24 | 30 | 40 | 40 | 64 | 70 | 8 | 22 | 28 | 38 | 38 | 64 | 70 |
| 005, 015 | X | X | FS11 | L2 | 100 | --- | 0.45 | 6 | 19 | 25 | 36 | 40 | 60 | 70 | 4 | 17 | 23 | 34 | 38 | 60 | 70 |
| 006, 016 | X | X | FS11 | C | 100 | --- | 0.45 | 6 | 19 | 25 | 36 | 40 | 60 | 70 | 4 | 17 | 23 | 34 | 38 | 60 | 70 |
| 007, 017 | X | X | FS11 | L2 | 150 | --- | 0.25 | --- | 14 | 20 | 31 | 40 | 56 | 70 | --- | 12 | 18 | 29 | 38 | 56 | 70 |
| 008, 018 | X | X | FS11 | C | 150 | --- | 0.25 | --- | 14 | 20 | 31 | 40 | 56 | 70 | --- | 12 | 18 | 29 | 38 | 56 | 70 |
| 009, 019 | X | X | FS11 | L2 | 200 | 125 | 0.15 | --- | 10 | 16 | 26 | 40 | 52 | 70 | --- | 9 | 14 | 25 | 38 | 52 | 70 |
| 010, 020 | X | X | FS11 | C | 200 | 125 | 0.15 | --- | 10 | 16 | 26 | 40 | 52 | 70 | --- | 9 | 14 | 25 | 38 | 52 | 70 |
| (C) 021, 031 | X | X | FS11 | L2 | 200 | 125 | 0.01 | ---- | ---- | ---- | 2 | 20 | 40 | 55 | ---- | ---- | ---- | ---- | 18 | 40 | 55 |
| (C) 022, 032 | X | X | FS11 | C | 200 | 125 | 0.01 | ---- | ---- | ---- | 2 | 20 | 40 | 55 | ---- | ---- | ---- | ---- | 18 | 40 | 55 |
| (C) 023, 033 | X | X | FS11 | L2 | 200 | 125 | 0.0027 | ---- | ---- | ---- | ---- | 10 | 28 | 45 | ---- | ---- | ---- | ---- | 8 | 28 | 45 |
| (C) 024, 034 | X | X | FS11 | C | 200 | 125 | 0.0027 | ---- | ---- | ---- | ---- | 10 | 28 | 45 | ---- | ---- | ---- | ---- | 8 | 28 | 45 |
| (C) 025, 035 | X | X | FS11 | L2 | 200 | 125 | 0.001 | ---- | ---- | ---- | ---- | 2 | 20 | 35 | ---- | ---- | ---- | ---- | 2 | 20 | 35 |
| (C) 026, 036 | X | X | FS11 | C | 200 | 125 | 0.001 | ---- | ---- | ---- | ---- | 2 | 20 | 35 | ---- | ---- | ---- | ---- | 2 | 20 | 35 |

1/ 0 to 400 Hz.
2/ Insertion loss measurements shall be made under full load over the frequency range of 150 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.
3/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.





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Marking: Filters and capacitors shall be marked, as a minimum, with the following information:

a. Military PIN.

b. JAN brand.

c. Source code.

d. Date code.

e. Terminal identification or circuit diagram (nonsymmetrical filters only).

In addition, full marking, in accordance with MIL-F-28861 shall be marked on the package.

Cataloging information. Dash numbers 002, 004, 006, 008, 010, 012, 014, 016, 018, 020, 022, 024, 026, 032, 034, and 036 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 003, 005, 007, 009, 011, 013, 015, 017, 019, 021, 023, 025, 031, 033, and 035 shall be cataloged under FSC 5915 as radio frequency interference filters.

## CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, SH, MC, OS
Air Force - 19, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0116)



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